

2303a52346 batch:45

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Ass 9.5

Problem 1: String Utilities Function

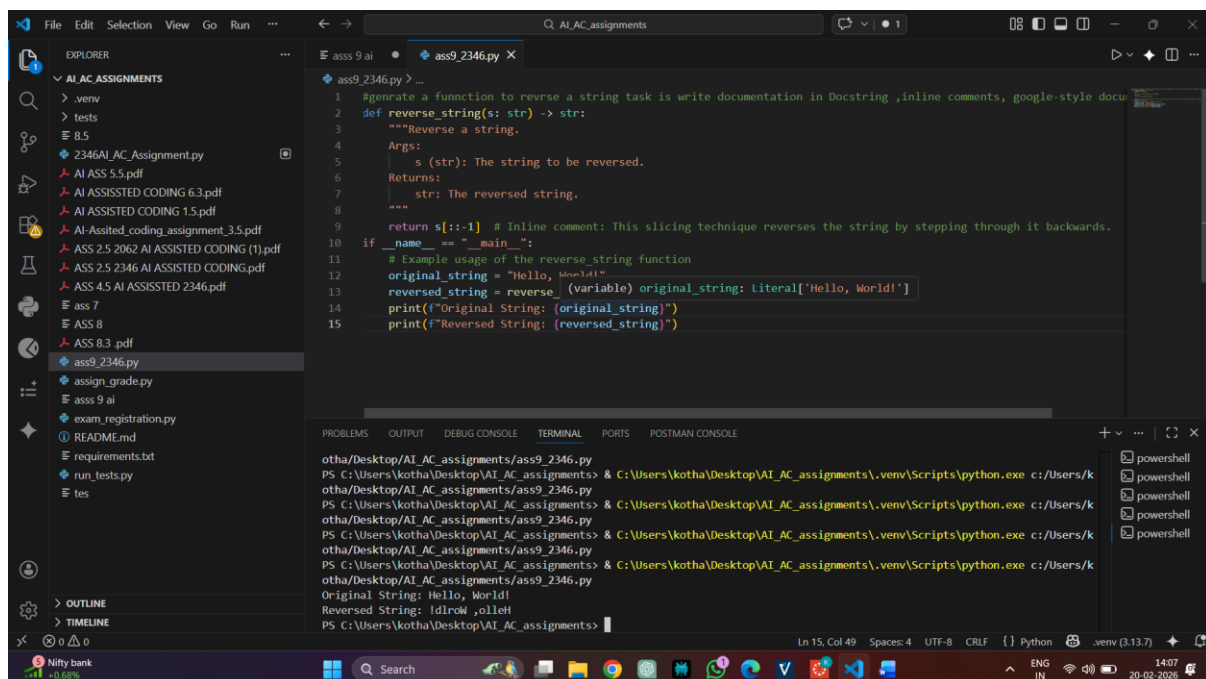
Consider the following Python function:

```
def reverse_string(text):  
    return text[::-1]
```

Task:

1. Write documentation in:
 - o (a) Docstring
 - o (b) Inline comments
 - o (c) Google-style documentation
2. Compare the three documentation styles.
3. Recommend the most suitable style for a utility-based string

Library



```
1 #generate a function to revrse a string task is write documentation in Docstring ,inline comments, google-style docu
2 def reverse_string(s: str) -> str:
3     """Reverse a string.
4     Args:
5         s (str): The string to be reversed.
6     Returns:
7         str: The reversed string.
8     """
9     return s[::-1] # Inline comment: This slicing technique reverses the string by stepping through it backwards.
10 if __name__ == "__main__":
11     # Example usage of the reverse string function
12     original_string = "Hello, World!"
13     reversed_string = reverse_string(original_string)
14     print(f"Original String: {original_string}")
15     print(f"Reversed String: {reversed_string}")
```

Terminal Output:

```
PS C:\Users\kotha\Desktop\AI_AC_assignments> python ass9_2346.py
Original String: Hello, World!
Reversed String: !dlroW ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments>
```

Problem 2: Password Strength Checker

Consider the function:

```
def check_strength(password):  
    return len(password) >= 8
```

Task:

1. Document the function using docstring, inline comments, and Google style.
2. Compare documentation styles for security-related code.
3. Recommend the most appropriate style.

```
38 #2
39 #Standard Docstring
40 def check_strength(password):
41     """
42     Checks whether a password meets minimum length requirements.
43
44     Parameters:
45     password (str): The password to validate.
46
47     Returns:
48     bool: True if password length is at least 8 characters.
49     """
50     return len(password) >= 8
51 #Inline Comments
52 def check_strength(password):
53     # Check if password length is at least 8 characters
54     return len(password) >= 8
55 #Google Style
56 def check_strength(password):
57     """
```

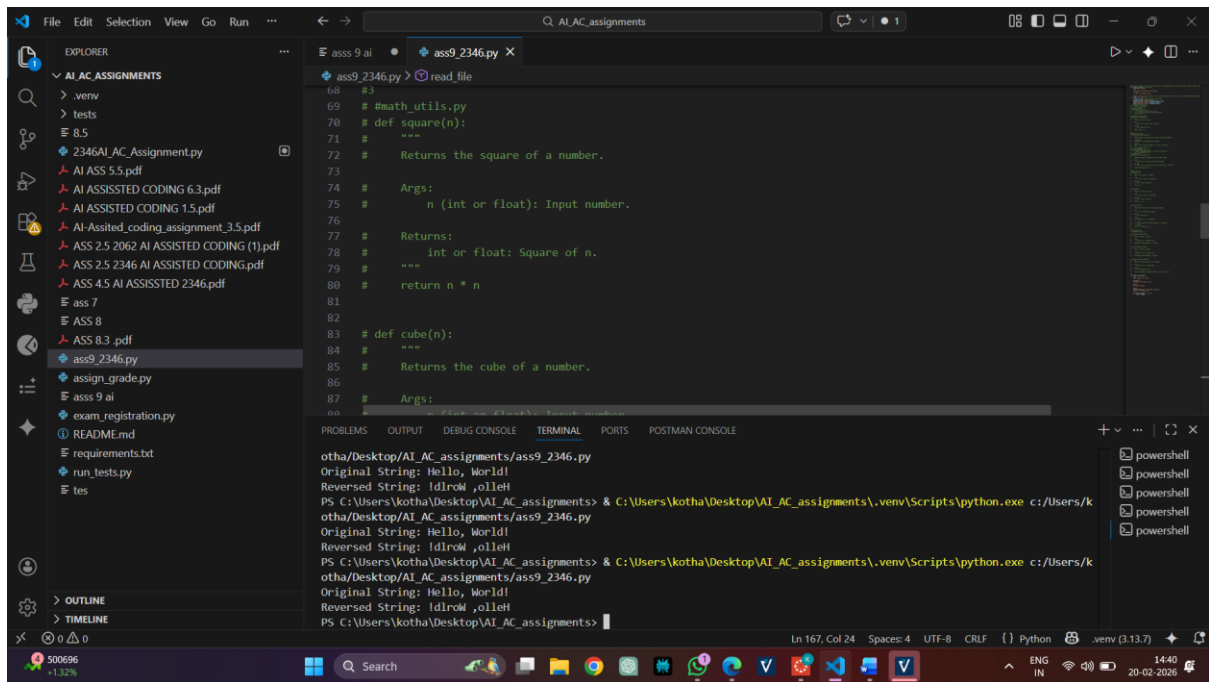
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe c:/Users/k
Original String: Hello, World!
Reversed String: !dlroW ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe c:/Users/k
Original String: Hello, World!
Reversed String: !dlroW ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe c:/Users/k
Original String: Hello, World!
Reversed String: !dlroW ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments>
```

Problem 3: Math Utilities Module

Task:

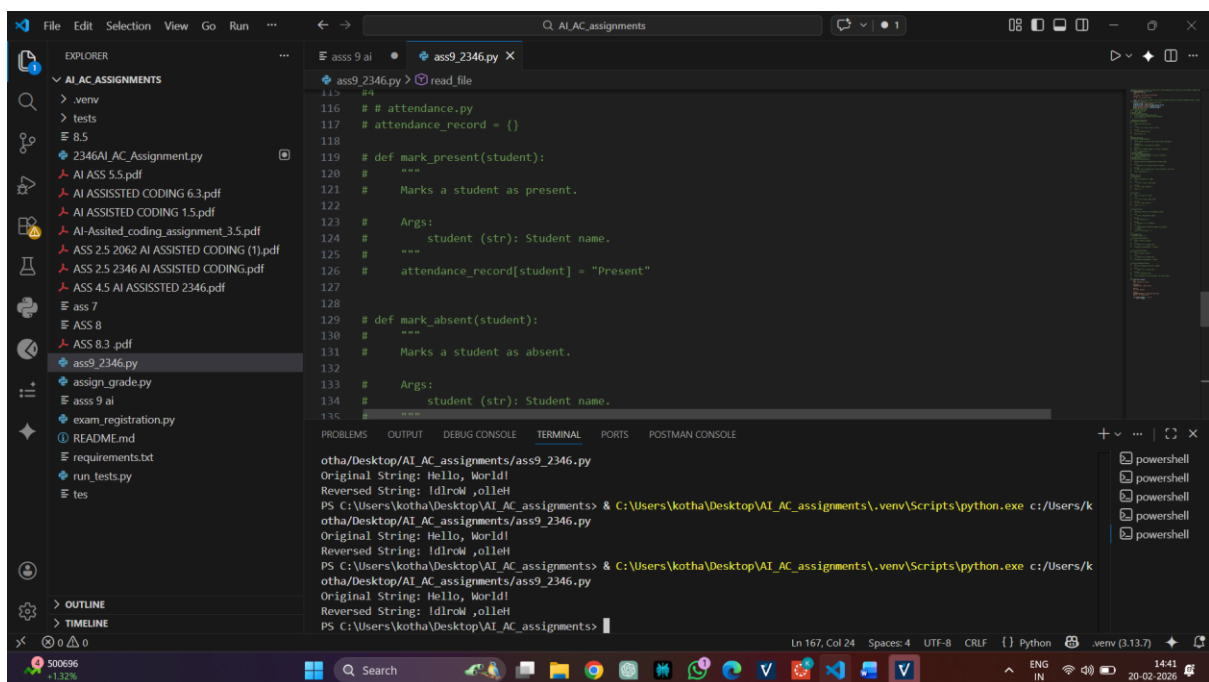
1. Create a module math_utils.py with functions:
 - o square(n)
 - o cube(n)
 - o factorial(n)
2. Generate docstrings automatically using AI tools.
3. Export documentation as an HTML file



Problem 4: Attendance Management Module

Task:

- Create a module `attendance.py` with functions:
 - `mark_present(student)`
 - `mark_absent(student)`
 - `get_attendance(student)`
- Add proper docstrings.
- Generate and view documentation in terminal and browse



Problem 5: File Handling Function

Consider the function:

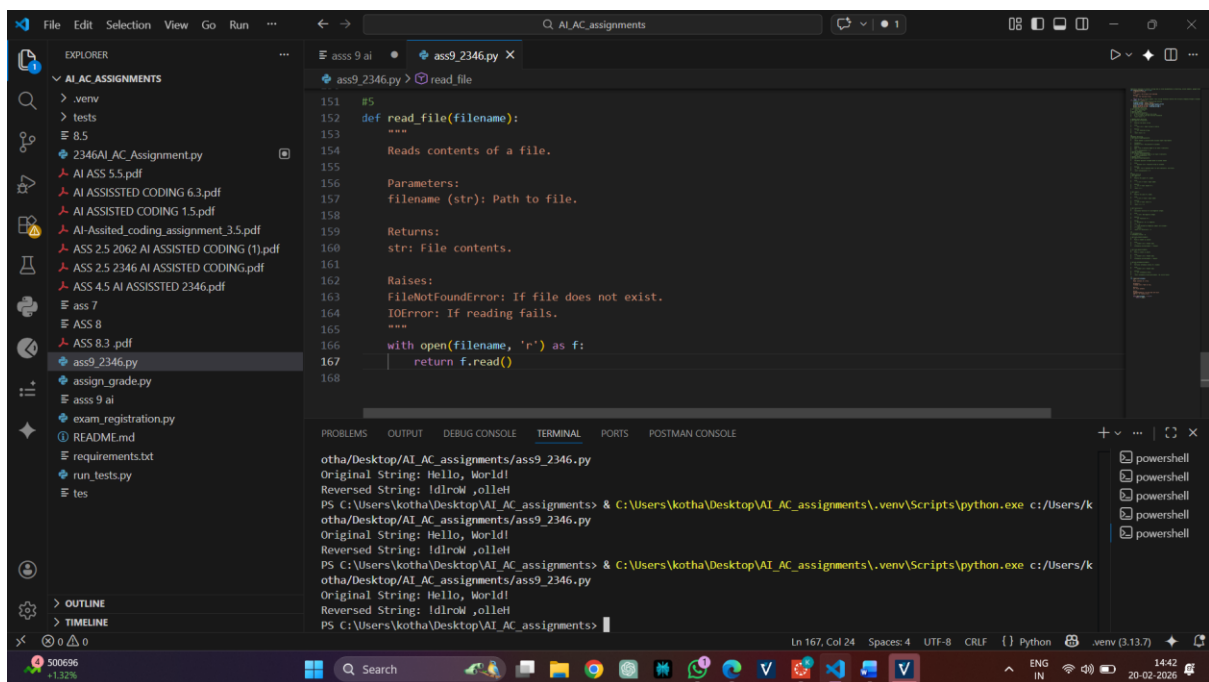
```
def read_file(filename):
```

```
    with open(filename, 'r') as f:
```

```
        return f.read()
```

Task:

1. Write documentation using all three formats.
2. Identify which style best explains exception handling.
3. Justify your recommendation.



```
151 #5
152 def read_file(filename):
153     """
154     Reads contents of a file.
155
156     Parameters:
157     filename (str): Path to file.
158
159     Returns:
160     str: File contents.
161
162     Raises:
163     FileNotFoundError: If file does not exist.
164     IOError: If reading fails.
165     """
166     with open(filename, 'r') as f:
167         return f.read()
168
```

ohta/Desktop/AI_AC_assignments/ass9_2346.py
Original String: Hello, World!
Reversed String: !dlröw ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe c:/Users/k
ohta/Desktop/AI_AC_assignments/ass9_2346.py
Original String: Hello, World!
Reversed String: !dlröw ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe c:/Users/k
ohta/Desktop/AI_AC_assignments/ass9_2346.py
Original String: Hello, World!
Reversed String: !dlröw ,olleH
PS C:\Users\kotha\Desktop\AI_AC_assignments>